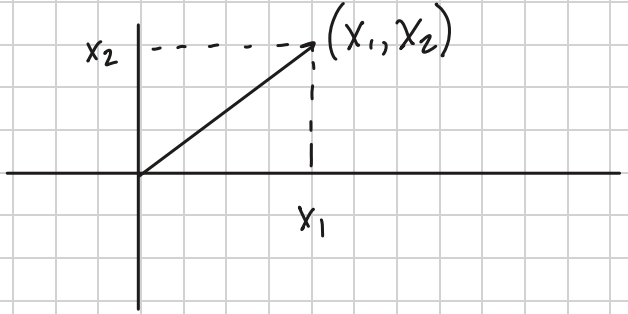
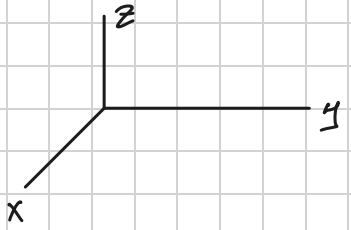
Week 7

$\mathbb{R}^2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$





Standard bases

$\mathbb{R}^2$ Standard basis vectors

$$\begin{bmatrix}1\\0\end{bmatrix} \qquad \begin{bmatrix}0\\1\end{bmatrix}$$

e_1 e_2

Standard

Standard

$$\begin{bmatrix}x_1\\x_2\end{bmatrix} = x_1 \cdot \begin{bmatrix}1\\0\end{bmatrix} + x_2 \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x_1\\0\end{bmatrix} + \begin{bmatrix}0\\x_2\end{bmatrix} = \begin{bmatrix}x_1\\x_2\end{bmatrix}$$

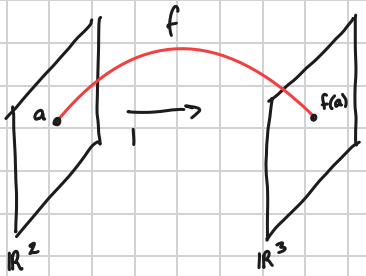
decompose or write in terms of linear comb of basis vectors

$$\begin{bmatrix}3\\1\end{bmatrix} = 3 \cdot e_1 + 1 \cdot e_2$$

Coefficients

$n=1$ $[x]$

+ define transformations/functions



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$